



AlgoRhythms
(NariY_11)

Tweet Analysis using Quantum

SDG 3: Good Health and Well being

Team Members: Tanisha Das, Swetha R, Swapnali
Singh
College: Christ University

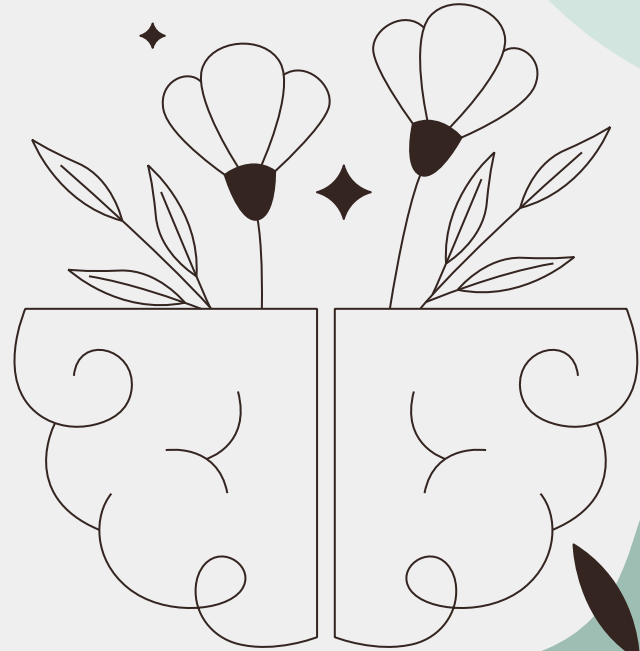


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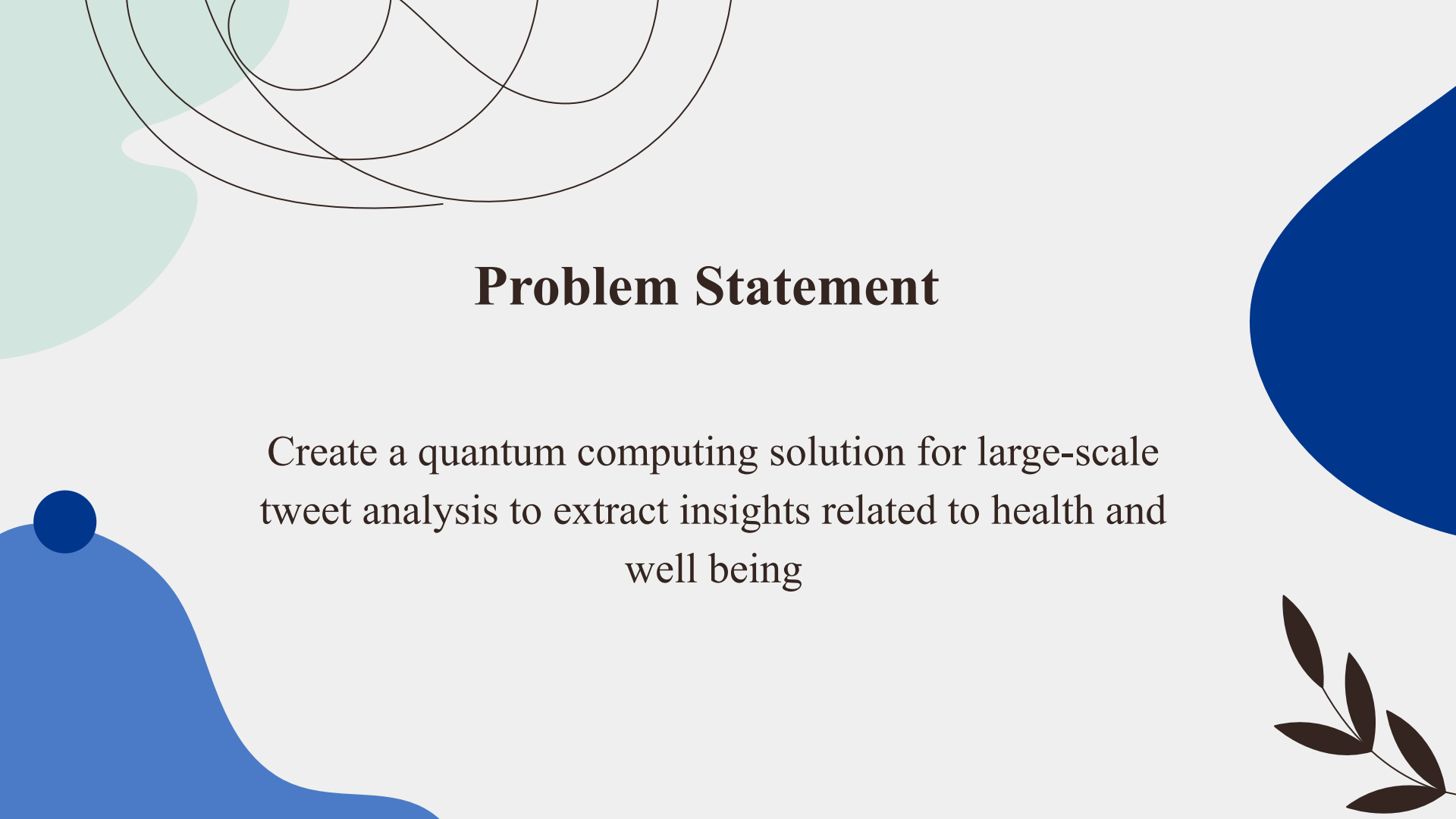
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Problem Statement

Create a quantum computing solution for large-scale tweet analysis to extract insights related to health and well being

Challenges:

- The vast volume of tweets makes manual filtering infeasible.
- Identifying health-related patterns or trends is time-intensive.
- Efficient analysis of sentiments related to public health is critical.

Goal:

To create an efficient **Quantum-powered solution** for large-scale tweet analysis to extract actionable insights on health and well-being.

Proposed Solution



Quantum Computing

Prioritize tweets containing health-related keywords using Grover's algorithm



NLP

Perform sentiment analysis on the filtered tweets to classify them as **'positive'**, **'negative'**.



Visualization Dashboard

Present insights in a user-friendly interface using charts and tables.

What is quantum Computing?

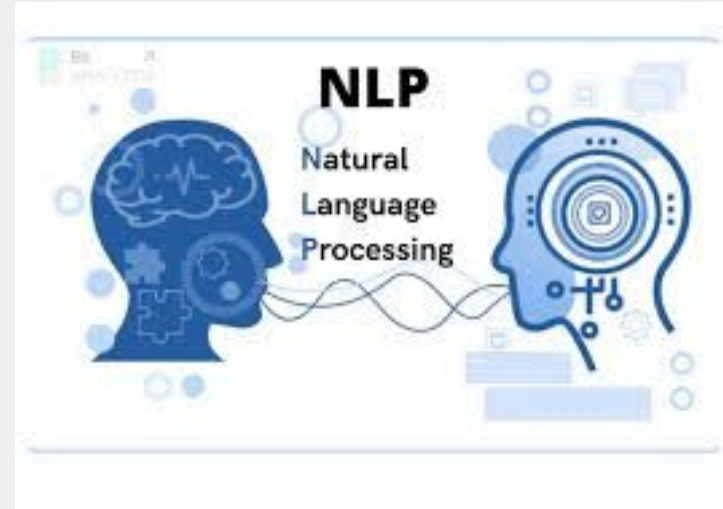
- Quantum computing is a multidisciplinary field comprising aspects of computer science, physics, and mathematics that utilizes quantum mechanics to solve complex problems faster than on classical computers.
- Some applications where quantum computers can provide such a speed boost include Machine Learning, optimization, and simulation of physical systems. Eventual use cases could be portfolio optimization in finance or the simulation of chemical systems, solving problems that are currently impossible for even the most powerful supercomputers on the market.

Example:

- **Traditional Computing:** This is like having a super-organized librarian who checks one book at a time. You want to find a specific book? The librarian walks up and down the shelves, checking each book one by one. Boring and time-consuming!
- **Quantum Computing:** This is like having a magical librarian who can read EVERY single book in the library at the EXACT SAME TIME. And not just read them - understand them, compare them, and find exactly what you need in a split second.

Natural Language processing:

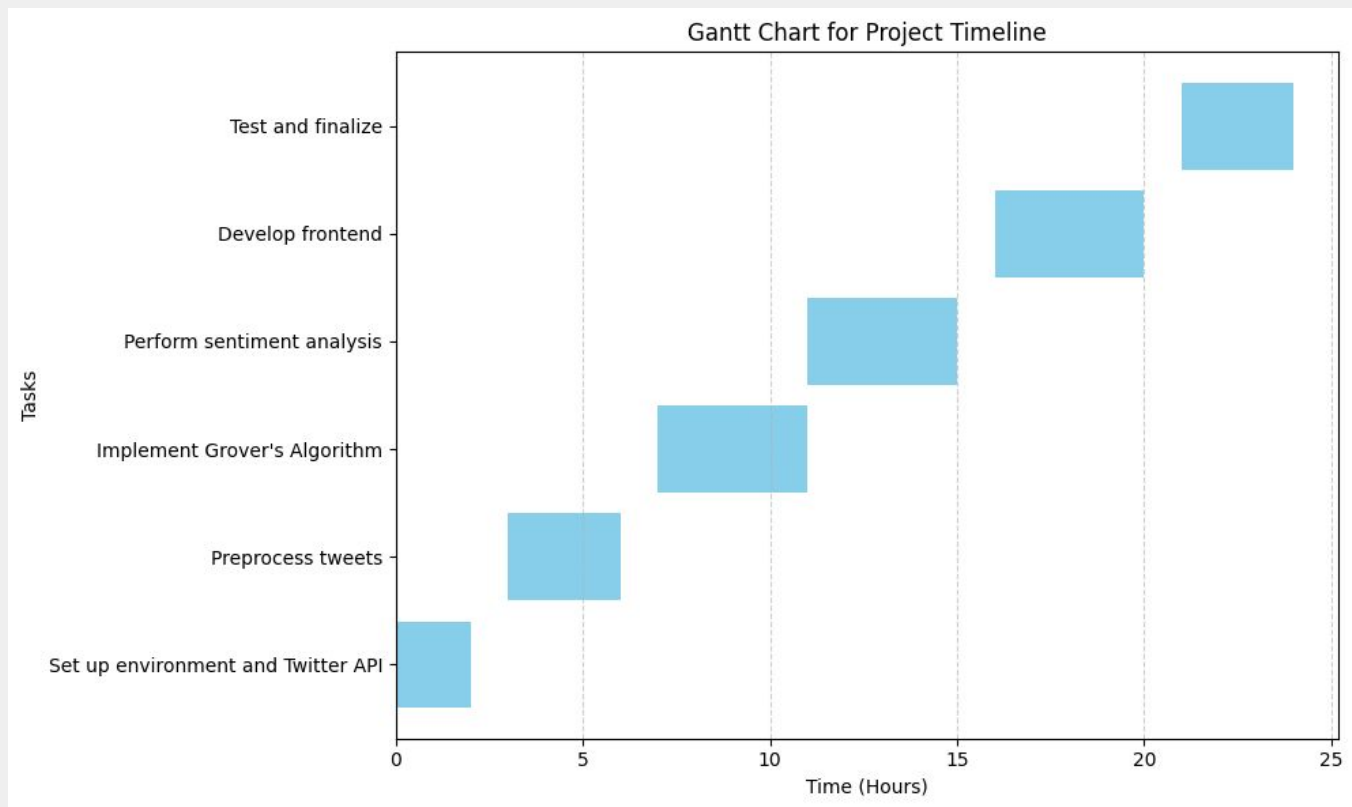
- Natural language processing (NLP) is a method computer programs can use to interpret human language. It can work through the differences in dialects, slang, and grammatical irregularities typical in day-to-day conversations.
- Organizations today have large volumes of voice and text data from various communication channels like emails, text messages, social media feeds, video, audio, and more. They use NLP software to automatically process this data, analyze the intent or sentiment in the message, and respond in real time to human communication.



Technical Stack

Component	Technology Used	Justification
Programming	Python	Widely used for data processing, quantum computing, and web development
Quantum	Qiskit	Industry-standard quantum framework for Grover's Algorithm and other quantum tasks.
NLP TextBlob	NLTK	Prebuilt tools for sentiment analysis and text preprocessing; beginner-friendly.
Frontend	Streamlit	Streamlit framework for building interactive dashboards;

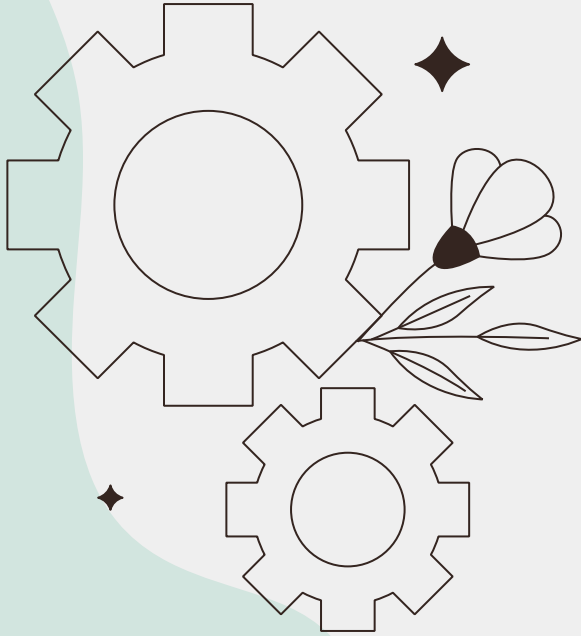
RoadMap



Feasibility

Implementation Feasibility:

- Each component is broken into manageable tasks, ensuring the solution is achievable within 24 hours.
- Prebuilt libraries like Qiskit and TextBlob streamline implementation.



Innovation And Creativity



Innovation:

- The project combines quantum computing with classical NLP, showcasing an innovative hybrid approach.
- Demonstrates quantum speedup in keyword search, a novel application in social media analytics.

Creativity:

- Develops a visually appealing, interactive dashboard for presenting insights.
- Explores future applications of quantum computing in large-scale public health analysis.

Impact

Impact on Society:

- Enables policymakers to monitor ‘**public sentiment on health issues**’ in real-time.
- Assists researchers in analyzing ‘**mental health trends**’, fitness patterns, and public health discussions.

Future Potential:

- Scaling to larger datasets with real quantum hardware.
- Expanding to other domains, like disaster response or political sentiment analysis.



Results:

Natural Language Processing(NLP)

tweet_data2.csv 60.8MB

Processing Time: 50.42 seconds

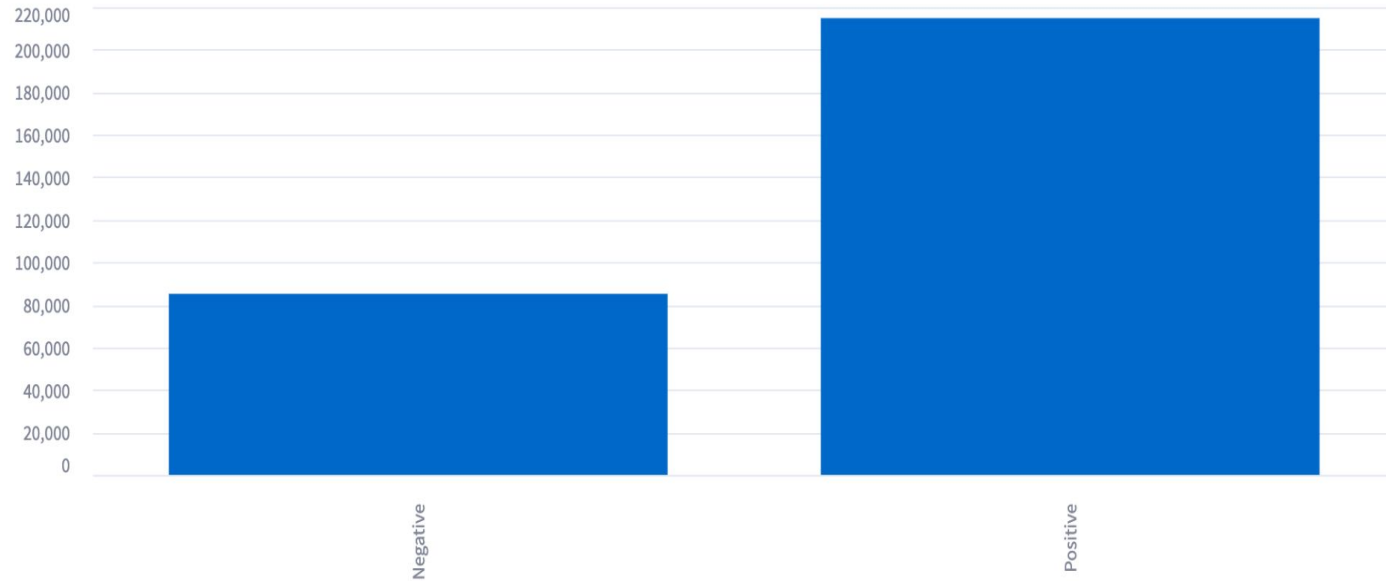
Processed Data

	user_name	date	text	cleaned_text
0	advosushildixit	2019-05-18 23:50:47+00:00	@anjanaomkashyap I am seeing you as future #bjp spokesperson.. Good luck. Anjana	@anjanaomkashyap i
1	airnewsalerts	2019-05-18 23:20:00+00:00	Trinamool Congress Sitting MP Abhishek Banerjee sends a defamation notice to Senior	trinamool congress sit
2	jiaeur	2019-05-18 23:00:03+00:00	#LokSabhaElections2019 23rd May 2019 will reveal .. Even ECISVEEP could not help M	#loksabhaelections20
3	PVenkatGandhi	2019-05-18 22:53:54+00:00	#LokSabhaElections2019 23rd May 2019 will reveal .. Even @ECISVEEP could not help	#loksabhaelections20
4	TheNirbhay1	2019-05-18 22:20:48+00:00	PM Modi creates a new record of being the only PM of a democratic country. Congrat	pm modi creates a nev
5	ankitgarg20	2019-05-18 22:16:26+00:00	My somewhat biased exit poll for India elections based on ground reports that I have	my somewhat biased e
6	pkm1999	2019-05-18 21:51:56+00:00	@rupasubramanya Even assuming statistical errors in the estimation , #BJP is expect	@rupasubramanya ev
7	rohanmodi	2019-05-18 21:43:15+00:00	@abhijitmajumder Small correction. Nobody gets appointed in #BJP as PM. They wo	@abhijitmajumder sm
8	Jmend124	2019-05-18 21:36:20+00:00	We still fucking dancing 🕺💃 #INC	we still fucking dancin
9	ShakeChilli	2019-05-18 21:22:29+00:00	@abhijitmajumder Appointment of Successor! God forbid if we allow BJP to stoop s	@abhijitmajumder ap

Natural Language Processing(NLP)



Sentiment Distribution



Quantum Search

Select an NLP Task

Choose a function:

- ☐ Home
- ☐ NLP Sentiment Analysis
- ☒ Quantum Search & Sentiment Analysis

Deploy

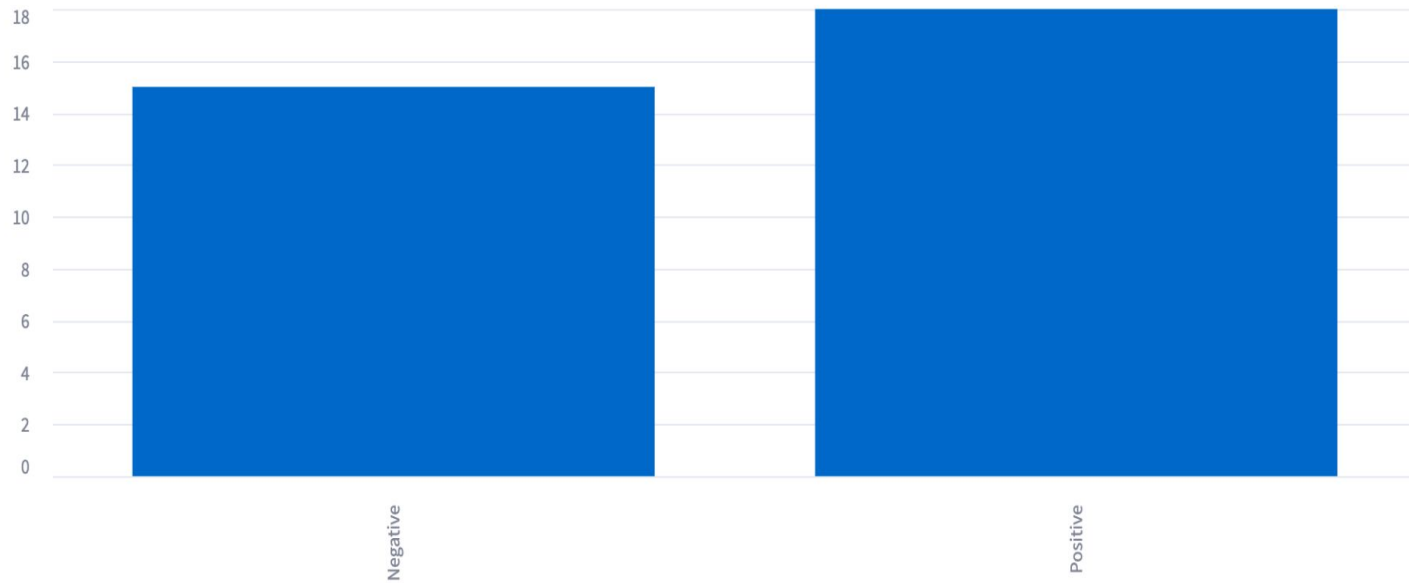
Analyzed tweets saved to analyzed_tweets.csv
Grover Pipeline Time: 0.01 seconds

Results

	Index	Tweet	Sentiment
0	12,131	#SquidGame is literally self sabotage for anyone with anxiety	Negative
1	13,140	If I didn't enjoy the Hunger Games would I also hate #SquidGame? Everyone's raving	Negative
2	16,092	because i fell in love with squid game, i drew a tiny sangwoo and a tiny gi hun. hope y	Positive
3	19,714	#SquidGame jun-ho is the best character. Such a joy to follow his side of the story, be	Positive
4	24,590	killionaire.music - squid game (cchekoz bootleg) https://t.co/1WMeo3bmqa I made	Positive
5	25,405	Best project ever! I am very happy to be participating in this project. I hope this proje	Positive
6	25,990	@rigelprotocol @AxiInfinity This is very potential project for future and i hope this p	Positive
7	29,119	I enjoyed #SquidGame but did NOT enjoy the ending 🙄	Negative
8	40,354	Signs of mental #strength and #resilience: https://t.co/6QaeGINVG7	Positive
9	41,502	Depression Treatments: I knew I was different long before I was diagnosed with Asper	Negative



Sentiment Distribution



Thank You